KENTECH 교육혁신 우수 사례: 통계분석 교과 프로젝트

성격 유형 별 반복 학습과 성취도 간 관계 분석

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연구 동기 및 목적

The motivation for this study stems from the widespread use of the Myers-Briggs Type Indicator (MBTI) and the need to explore the validity of its practical application. By providing a minimum population in which all MBTIs can be present, it provides practicality for future MBTI-related research and allows us to understand the relationship between MBTI personality types and cognitive performance through N (intuition) and S (sensing) preferences. Ultimately, this study aims to contribute to the understanding and practical application of the MBTI assessment while recognizing the need for further research and analysis to establish a comprehensive understanding of personality type and its impact on cognitive performance.

배경 이론

The **MBTI**[1] is an introspective self-report questionnaire indicating differing psychological preferences in how people perceive the world and make decisions.

N Intuitives prefer to focus on possibilities and the big picture, easily see patterns, value innovation and seek creative solutions to problems.

Sensors are realistic people who like to focus on the facts and details, and apply common sense and past experience to come up with practical solutions to problems. S

Sensory training[2] refers to specific practices designed to enhance and refine the functions of our sensory systems, which include sight (vision), hearing (auditory), touch (tactile), smell (olfactory), taste (gustatory), balance (vestibular), and body awareness (proprioception). Also, Sensory training is underpinned by a few core scientific principles which is neuroplasticity, Hebbian theory, and the principles of learning and memory consolidation.

Math methods are needed for analysis result of experiment.

- Mean $\mu = (x_1 + x_2 + \cdots + x_n)/n$
- Variance $\sigma^2 = [(x_1 - \mu)^2 + (x_2 - \mu)^2 + \cdots + (x_n - \mu)^2]/n$
- Normal Distribution is continuous probability distribution for a random variable

$$f(x) = \left(\frac{1}{\sqrt{2\pi\sigma^2}}\right)e^{\left(\frac{-(x-\mu)^2}{2\sigma^2}\right)}$$

$f(x)$ is the probability density function
x is any point on the continuous number line
μ is the mean of the distribution
σ is the standard deviation (the square root of variance)
e is the base of natural logarithm, approximately equal to 2.71828
π is a mathematical constant, approximately equal to 3.14159

연구 방법

Study 1: Find the minimum number of groups that can include every type of MBTI

Based on data of South Korea MBTI result population ratio, we analyzed the percentage per each MBTI, and the sample size was 70,266. According to the data, we found the number of people with the 'lowest percentage MBTI.' Then we multiplied the total number of samples to obtain the smallest number of MBTI, then divide it to obtain the minimum number of groups for the trial.

Study 2: Experiment on the relationship between sensory perception and MBTI

Based on result of Study 1, we conducted Study 2 with 52 students from KENTECH. For the experiment, Participants' eyes are closed, and the moment they intuitively feel that 10 seconds have passed is measured. Participants' MBTI types are also recorded. The collected data is analyzed to determine if there is a significant relationship with MBTI types (N, S Intuition Sensing). The experiment was conducted on 52 randomly selected students from the first and second grades of KENTECH. Four of us ran the experiment for a week, organizing and sharing our progress in Excel file.

Study 3: Experiment on the relationship between repeated trials and sensory learning

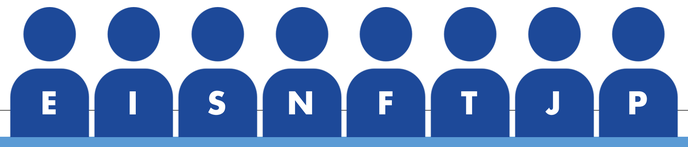
Randomly select one student per MBTI (16 students) and repeat the same experiment as Study 2. This time, instead of just running it once, repeat the same experiment until the error in the results is less than 0.1 seconds. For each trial, record the results, record the number of times the error is less than 0.1 seconds in 10 seconds measured with eyes closed, and stop. Based on the participants' experiments, it is inferred that sensory learning occurs correctly after a minimum of n attempts.

참조

[1] The Myers&Briggs Foundation https://www.myersbriggs.org/my-mbti-personality-type/

[2] Hebb, D.O. (1949). The Organization of Behavior. New York: Wiley & Sons.

[3] Republic of Korea Personality Profile – 16 personalities
https://www.16personalities.com/country-profiles/republic-of-korea



연구 결과

Study 1

[3](Total Sample Size)*(Lowest Percentage=ENTJ)

70,266*0.0273 = 1,918.2618

(Total Sample Size)/(Size of ENTJ)

70266/1,918.2618 = 36.6300

Minimum number of groups: **37**

	Name	MBTI	10s	Error	N/S		Name	MBTI	10s	Error	N/S
1	정O윤	ENFJ	8.98	1.02	N	27	최O호	ISTP	7.82	2.18	S
2	민O윤	INTJ	7.86	2.14	N	28	김O우	ESTJ	8.55	1.45	S
3	강O의	ISTJ	10.45	0.45	S	29	방O영	ESTP	8.79	1.21	S
4	김O민	ENTP	9.19	0.81	N	30	김O홍	ISFJ	9.9	0.1	S
5	박O준	INTJ	9.29	0.71	N	31	김O성	ISFP	8.93	1.07	S
6	이O하	ENFP	8.07	1.93	N	32	조O호	INFP	9.67	0.33	N
7	황O성	INTP	8.93	1.07	N	33	이O연	INTP	9.79	0.21	N
8	노O호	ISTP	9.77	0.23	N	34	김O석	ENFP	11.64	1.64	N
9	오O윤	ISTJ	9.95	0.05	S	35	방O빈	ENFJ	7.89	2.11	N
10	정O민	ISTP	7.86	2.14	S	36	이O희	ESTJ	11.52	1.52	S
11	김O	ISFJ	9.68	0.32	N	37	채O민	ENFP	7.91	2.09	N
12	정O훈	INFP	8.51	1.49	N	38	선O원	INFP	6.82	3.18	N
13	최O우	ISFJ	9.82	0.18	S	39	이O건	INTP	11.23	1.23	N
14	한O진	ESTJ	8.39	1.61	S	40	선O류	ESFJ	10.12	0.12	S
15	이O우	INFP	10.01	0.01	N	41	김O원	INTP	11.6	1.6	N
16	이O희	ENFP	8.12	1.88	N	42	황O원	ENTP	9.9	0.1	N
17	최O정	ENFP	9.92	0.08	N	43	박O영	INTJ	9.98	0.02	N
18	노O의	INTJ	10.34	0.34	N	44	최O혁	ISTP	8.35	1.65	S
19	이O영	ENFP	8.56	1.44	N	45	손O준	ENFP	8.28	1.72	N
20	이O진	INTJ	10.07	0.07	N	46	강O우	ENTP	8.55	1.45	N
21	이O민	ISFJ	9.34	0.66	S	47	배O규	INTP	10.6	0.6	N
22	서O서	ESFP	12.37	2.37	S	48	김O일	ENFP	10.37	0.37	N
23	강O수	ENTP	11.24	1.24	N	49	신O용	INTP	10.51	0.51	N
24	김O현	ENFP	8.59	1.41	N	50	김O빈	ISTP	10.57	0.57	S
25	조O일	ESFJ	9.88	0.12	S	51	김O영	ISTP	12.22	2.22	S
26	김O준	ENTJ	7.69	2.31	N	52	정O원	INTP	10.99	0.99	N

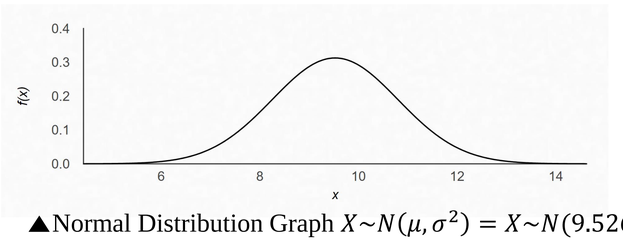
Study 2

Mean of Error $E(R) = \frac{\sum(r)}{52} = 1.083$ *(r=measured error[s])*

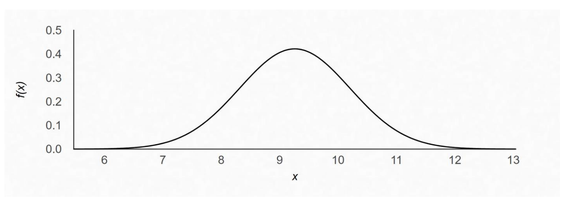
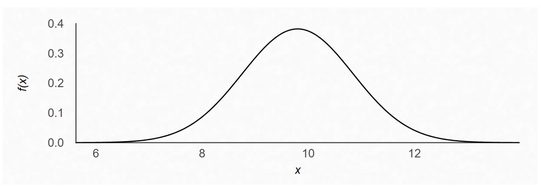
Mean of Self-measured 10 seconds $E(X) = \frac{\sum(x)}{52} = 9.526$ *(x=measured time[s])*

Variance of Self-measured 10 seconds $V(X) = \frac{\sum_{i=1}^{52}(X_i - 9.526)^2}{52} = 1.625$

Deviation of Self-measured 10 seconds $\sigma(X) = \sqrt{V(X)} = 1.275$



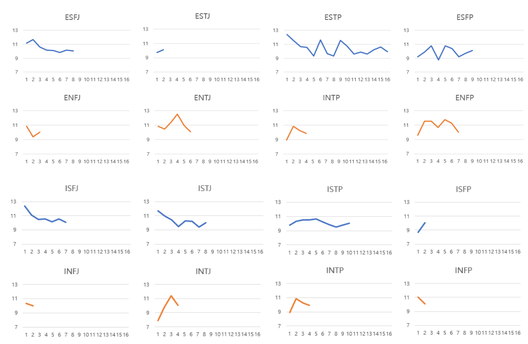
▲Normal Distribution Graph $X \sim N(\mu, \sigma^2) = X \sim N(9.526, 1.275)$

N MBTI with N (32)	S MBTI with S (20)
Mean $E(X) = \frac{\sum(x)}{32} = \mathbf{9.26}$ *(x=measured time[s])* Variance $V(X) = \frac{\sum_{i=1}^{32}(X_i - 9.26)^2}{32} = 0.895$ Deviation $\sigma(X) = \sqrt{V(X)} = 0.946$	Mean $E(X) = \frac{\sum(x)}{20} = \mathbf{9.80}$ *(x=measured time[s])* Variance $V(X) = \frac{\sum_{i=1}^{20}(X_i - 9.8)^2}{20} = 1.091$ Deviation $\sigma(X) = \sqrt{V(X)} = 1.044$
	

We can interpret that MBTI with S are more accurate than MBTIs with N when measuring in a sensory way, using their body's senses.

Study 3

	N	S
Average initial error (s)	0.0237	0.62625
Average final error (s)	0.01375	0.045
Average required trials	4.25	7.75
Variance required trials	3.1875	17.6875
Standard Deviation required trials	1.785357	4.205651



▲Graph showing data change of trial by MBTI

We can interpret that initial average error of MBTI with N tends to be smaller than MBTI with S which means that MBTI with N tends to be more sensitive that S. Also, we can conclude that repeated behavior shows high learning efficiency when people have MBTI with N.

결론 및 고찰

The results of this study provide insight into the correlation between MBTI personality types and performance on a 10-second estimation task. The results showed that people with an N (intuition) preference needed fewer trials to perform the 10-second estimation task. Based on these findings, we can predict that people with an N preference can reinforce and fix concepts through repetition-based learning, while people with an S preference are efficient at connecting new information with existing knowledge through insight-based learning. This suggests that the results of the experiment may have utility.

Furthermore, while this study suggests a minimum population in which all MBTI types can exist based on their statistical distribution, it is important to recognize that MBTI types may not actually be distributed in a uniform proportion. Assuming an equal distribution without considering a variety of variables may limit the external validity of the results.

Finally, the ability to complete a 10-second estimation task is not a comprehensive representation of human cognitive ability or intelligence. It is only one of the specific tasks in this study. Additional research is needed to explore the relationship between more diverse cognitive tasks or abilities across MBTI personality types.